

# A bi-Lipschitz class missed by every sublinear displacement class

Solution packet for arXiv:2102.13046

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**Status.** Claimed full classification of the source’s intersection question; likely valid pending human review.

## 1 Source question

Dymond and Kaluža introduce  $\phi$ -displacement equivalence of separated nets in  $\mathbb{R}^d$  [2]. If  $h : X \rightarrow Y$  is a bijection, write

$$\text{disp}_R(h) = \sup\{|h(x) - x| : x \in X, |x| \leq R\}.$$

The nets are  $\phi$ -displacement equivalent if some bijection has  $\text{disp}_R(h) = O(\phi(R))$ . After proving that the  $\phi$ -class of  $\mathbb{Z}^d$  meets uncountably many bi-Lipschitz (BL) classes for every unbounded, increasing, concave  $\phi$ , they ask:

Does every such  $\phi$ -displacement equivalence class intersect every BL equivalence class?

Indeed the ‘only if’ implication of Conjecture 1.8 for the modulus of continuity  $\omega$  of (1) is precisely the result [3, Proposition 1.3] referred to above. In Section 6 of the present work, we show that for every increasing, unbounded, concave function  $\phi : (0, \infty) \rightarrow (0, \infty)$ , the  $\phi$ -displacement class of the integer lattice intersects distinct BL classes; in particular it contains  $\omega$ -irregular separated nets for  $\omega(t) = t$ . A matter of interest is whether every such  $\phi$ -displacement equivalence class intersects every BL equivalence class. This question remains open, but we are able to show that every such  $\phi$ -displacement equivalence class intersects uncountably many BL equivalence classes:

The answer is negative for every sublinear  $\phi$ , and positive exactly at linear scale.

## 2 Idea of the proof

Burago and Kleiner constructed a bounded positive density  $\rho$  on a cube that is not the Jacobian density of any bi-Lipschitz map [1]. We build a separated net  $X$  whose normalized counting measures on rapidly growing cubes converge to  $\rho dx$ .

Suppose a BL image  $Y$  of  $X$  could be matched to  $\mathbb{Z}^d$  with sublinear displacement. Blow up the BL map on those cubes. Compactness produces a bi-Lipschitz limit  $F$ . The sublinear matching disappears under rescaling, so the images of the sampled points become the full lattice inside  $F(Q)$ , up to a vanishing boundary layer. Hence

$$F_{\#}(\rho dx) = dx|_{F(Q)},$$

contradicting the defining property of  $\rho$ . Importantly, the net  $X$  is independent of  $\phi$ , so one BL class is missed simultaneously by all sublinear lattice-displacement classes.

### 3 A net with a prescribed blow-up density

A *separated net* (or Delone set) is a subset of  $\mathbb{R}^d$  with positive minimum separation and finite covering radius.

Fix  $d \geq 2$  and  $Q = [0, 1]^d$ . By the Burago–Kleiner obstruction, in the form used in the proof of [2, Theorem 6.1], there is a measurable  $\rho : Q \rightarrow (0, \infty)$  with  $\int_Q \rho \, dx = 1$  and

$$0 < m \leq \rho \leq M < \infty \quad (1)$$

such that no bi-Lipschitz  $F : Q \rightarrow \mathbb{R}^d$  satisfies

$$F_{\#}(\rho \, dx) = dx|_{F(Q)}. \quad (2)$$

The normalization is harmless: multiplying a density by a positive constant is compensated by a scalar dilation of the target, so realizability is unchanged.

**Lemma 1** (Nested-shell sampling). *There are a separated net  $X \subset \mathbb{R}^d$  and integers  $R_n \rightarrow \infty$  such that*

$$\mu_n := \frac{1}{R_n^d} \sum_{x \in X \cap R_n Q} \delta_{x/R_n} \xrightarrow{\text{weakly}} \rho \, dx|_Q. \quad (3)$$

*Proof.* Choose  $R_n$  so fast that  $R_{n-1}/R_n \rightarrow 0$ . The finite sampling lemma [2, Lemma 6.4] gives, for each  $n$ , a uniformly separated and uniformly relatively dense set  $\Xi_n \subset R_n Q$  whose normalized counting measure, after the dilation  $x \mapsto x/R_n$ , tends weakly to  $\rho \, dx$ . The separation and covering constants depend only on the two bounds in (1).

On the shell  $R_n Q \setminus R_{n-1} Q$ , use  $\Xi_n$ . Near every interface, delete points in a fixed-width strip and fill a slightly wider strip with a fixed translated lattice. The widths can be chosen once, from the uniform separation and covering constants, so the union over all shells is a separated net  $X$ .

At scale  $R_n$ , replacing  $\Xi_n$  by  $X$  changes the normalized measure only inside the old cube  $R_{n-1} Q$  and fixed-width interface strips. Uniform point-count bounds for Delone sets show that their masses are respectively

$$O((R_{n-1}/R_n)^d) \quad \text{and} \quad O(R_n^{-1}).$$

Both tend to zero. The weak limit is therefore unchanged, proving (3).  $\square$

### 4 The blow-up obstruction

**Lemma 2** (Bi-Lipschitz blow-up and lattice filling). *Let  $X$  and  $R_n$  satisfy (3). Suppose that a separated net  $Y$  admits an  $L$ -bi-Lipschitz bijection  $f : X \rightarrow Y$  and that a bijection  $h : Y \rightarrow \mathbb{Z}^d$  satisfies*

$$\text{disp}_R(h) = o(R). \quad (4)$$

*Then, after passing to a subsequence, there is an  $L$ -bi-Lipschitz map  $F : Q \rightarrow \mathbb{R}^d$  satisfying (2).*

*Proof.* Fix  $x_0 \in X$  and put  $b = f(x_0)$ . On  $R_n^{-1} X$  define

$$F_n(x/R_n) = \frac{f(x) - b}{R_n}, \quad G_n(x/R_n) = \frac{h(f(x)) - b}{R_n}. \quad (5)$$

**Compactness.** On every fixed cube  $P$ , the domains  $R_n^{-1} X \cap P$  become dense and the  $F_n$  are uniformly  $L$ -bi-Lipschitz. Extend them to  $P$  by Kirszbraun's theorem. Arzelà–Ascoli and a diagonal subsequence give local uniform convergence to an  $L$ -Lipschitz map  $F$  on  $\mathbb{R}^d$  (a neighborhood of  $Q$  would suffice). To pass the lower bound to the limit, approximate any two points of  $P$  by points of  $R_n^{-1} X$  and use the upper Lipschitz bound of the extensions. Thus

$$L^{-1}|u - v| \leq |F(u) - F(v)| \leq L|u - v|, \quad (6)$$

so  $F$  is bi-Lipschitz.

For  $x/R_n$  in a fixed  $P$ , bi-Lipschitzness of  $f$  gives  $|f(x)| = O(R_n)$ . Consequently (4) gives, uniformly on such points,

$$|G_n(x/R_n) - F_n(x/R_n)| = \frac{|h(f(x)) - f(x)|}{R_n} \rightarrow 0. \quad (7)$$

Pushing (3) forward and using local uniform convergence therefore yields

$$\nu_n := \frac{1}{R_n^d} \sum_{x \in X \cap R_n Q} \delta_{(h(f(x)) - b)/R_n} \rightharpoonup F_{\#}(\rho \, dx|_Q). \quad (8)$$

**Lattice filling.** Every atom of  $\nu_n$  lies in the shifted lattice

$$\mathcal{L}_n := R_n^{-1}(\mathbb{Z}^d - b),$$

whose normalized counting measures converge to Lebesgue measure. We show that the atom set of  $\nu_n$  differs from  $\mathcal{L}_n \cap F(Q)$  only near  $\partial F(Q)$ .

First, (7) and uniform convergence imply that images of points in  $R_n Q$  lie in an  $o(1)$ -neighborhood of  $F(Q)$ .

Conversely, suppose  $z_n \in \mathcal{L}_n$  stays in a compact subset of  $\text{int } F(Q)$ . By bijectivity of  $h \circ f$ , its unscaled version has the unique form  $h(f(x_n)) - b$ . Since that point has norm  $O(R_n)$ , sublinearity forces

$$|f(x_n)| = O(R_n), \quad |h(f(x_n)) - f(x_n)| = o(R_n). \quad (9)$$

Indeed, if  $|f(x_n)|/R_n \rightarrow \infty$ , then  $|h(f(x_n)) - f(x_n)| = o(|f(x_n)|)$  would imply  $|h(f(x_n))| \sim |f(x_n)|$ , a contradiction. The lower Lipschitz bound for  $f$  now gives  $|x_n| = O(R_n)$ .

After taking a subsequence,  $x_n/R_n \rightarrow t$  in a fixed compact cube, and (9) gives

$$F(t) = \lim z_n \in \text{int } F(Q).$$

Because  $F$  is a homeomorphism on a neighborhood of  $Q$ , invariance of domain gives  $F(\text{int } Q) = \text{int } F(Q)$ . Injectivity therefore forces  $t \in \text{int } Q$ , and hence  $x_n \in R_n Q$  eventually. By compactness, this conclusion is uniform on every compact subset of  $\text{int } F(Q)$ .

It follows that the symmetric difference in question is contained in a shrinking neighborhood of  $\partial F(Q)$ . Since  $\partial F(Q) = F(\partial Q)$  and  $F$  is Lipschitz, this boundary has Lebesgue  $d$ -measure zero. Counting unit cubes around points of the shifted lattice shows that the normalized lattice mass of the shrinking boundary layer tends to zero. Therefore

$$\nu_n \rightharpoonup dx|_{F(Q)}. \quad (10)$$

Comparing (8) and (10) proves (2).  $\square$

## 5 Full classification

**Theorem 1.** *For every  $d \geq 2$  there is a separated net  $X \subset \mathbb{R}^d$  such that no net BL-equivalent to  $X$  admits a sublinear-displacement bijection to  $\mathbb{Z}^d$ . The same  $X$  works for every sublinear displacement scale.*

*Proof.* Choose the non-realizable  $\rho$  and the net  $X$  from Lemma 1. If a BL-equivalent net  $Y$  admitted a bijection  $h : Y \rightarrow \mathbb{Z}^d$  satisfying (4), Lemma 2 would produce a bi-Lipschitz solution of (2), contrary to the choice of  $\rho$ .  $\square$

**Corollary 1.** *Let  $d \geq 2$  and let  $\phi : (0, \infty) \rightarrow (0, \infty)$  be increasing and concave. Every  $\phi$ -displacement class intersects every BL class if and only if  $\phi(R) \in \Omega(R)$ .*

*Proof.* If  $\phi = o(R)$ , Theorem 1 says that the  $\phi$ -class of  $\mathbb{Z}^d$  misses the BL class of  $X$ . If  $\phi \in \Omega(R)$ , source Proposition 2.6 gives an  $O(R)$ -displacement bijection between any two separated nets, so  $\phi$ -equivalence is indiscrete. These cases exhaust increasing concave  $\phi$ : their asymptotic slope exists.  $\square$

In dimension one every separated net is BL-equivalent to  $\mathbb{Z}$ , so the  $d \geq 2$  hypothesis is essential.

## 6 Verification, upgrades, and novelty

Eight focused attempts are recorded in the accompanying attempt note. The main upgrades were from an ordinary density mismatch to a non-realizable density, from scale-dependent far-away blocks to one global nested-shell net, and from pointwise image containment to the two-sided boundary-filling argument in Lemma 2. No computational assertion is used.

The published 2024 source still labels the question open. A bounded search on 11 August 2026 used the exact question, title, DOI, author names, and displacement/BL-class terminology. The later Dymond–Kaluža papers located concern bi-Lipschitz extension, not displacement-class intersections. No later solution was found. Novelty confidence is moderate pending a complete citation audit.

**Human-review recommendation.** Check Lemma 1’s fixed-width shell gluing, the lower bi-Lipschitz limit in (6), and especially the reverse inclusion in the lattice-filling paragraph. If these pass, Corollary 1 fully answers the stated question.

## References

- [1] D. Burago and B. Kleiner, *Separated nets in Euclidean space and Jacobians of biLipschitz maps*, Geom. Funct. Anal. 8, 273–282 (1998), doi:10.1007/s000390050056.
- [2] M. Dymond and V. Kaluža, *Divergence of separated nets with respect to displacement equivalence*, Geom. Dedicata 218, article 15 (2024), arXiv:2102.13046, doi:10.1007/s10711-023-00862-3.